22AIE463 TIME SERIES ANALYSIS

Predictive Modeling of Non-Stationary Tyre Degradation - A Time Series Analysis Approach to Formula 1 Race Strategy

PRESENTED BY: TEAM A9

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INTRODUCTION

The Critical Role of Tyres in F1 Performance

Tyres are the sole contact point between the car and the track, making them absolutely crucial for achieving optimal grip, speed, and overall performance in Formula 1 racing. Their condition directly dictates a car's handling and competitive edge.

Tyre degradation significantly impacts lap times and race pace, forcing teams to make critical strategic decisions. How a tyre wears down can determine whether a driver gains or loses positions.

Modern F1 races are a delicate balance of managing tyre wear, optimizing their lifespan, and executing strategically timed pit stops. These elements collectively dictate the outcome of a Grand Prix.

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Problem Statement

Formula 1 Pit Stop Timing Optimization

Tyre degradation plays a critical role in determining race performance and pit stop strategy in Formula 1. The condition of a tyre changes dynamically over time due to multiple factors such as

- track temperature
- fuel load
- driving style
- tyre compound

This degradation behavior is non-stationary, meaning its statistical properties change across race laps and stints.

Can we accurately model and forecast this non-stationary tyre degradation using time series analysis to enable optimal pit stop and race strategy decisions?

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KEY RESEARCH

Literature Review: Tire Degradation and Pit Stop Strategies

SNo.	Paper Title/Author	Year	Dataset	Methodology	Remark
1	Explainable Time Series Prediction of Tyre Energy in Formula One Race Strategy Jamie Todd, Junqi Jiang	2025	Serialised and tabular raw data by the Mercedes-AMG PETRONAS F1 Team, taken from real-time sources such as the car telemetry and GPS	Multivariate time series forecasting using: Linear Regression, XGBoost, RNN, LSTM, GRU, TFT; plus Explainable AI (feature importance + counterfactual explanations)	Strong predictive model for tyre energy; includes explainability; does not directly optimize pit stop decisions.
2	A State-Space Approach to Modeling Tire Degradation in Formula 1 Racing C. Cappello & S. Hoegh	2025	Publicly available Formula 1 timing data from FastF1 Python API	Bayesian state-space modelling: latent tyre degradation estimated with pit stops as state resets and lap time modelled as function of fuel and latent pace.	Probabilistic modelling captures latent tyre dynamics; adds uncertainty estimates; focuses on modelling, not direct strategy optimization.
3	Real Time Tyre Strategy Decision in F1 Using A* Algorithm Rinaldi M., et al	2025	Simulation-like model states (lap, tyre compound, wear level, track conditions); no large real dataset — heuristic/transition environment for A* evaluation	Search-based optimization (A*): cost function for lap time & wear, heuristic estimating future performance; finds optimal pit sequence minimizing total race time.	Algorithmic decision model for pit strategy; does not use time series prediction directly; depends on heuristic cost estimates.

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SUMMARY

Literature Summary

Time Series Prediction Models use multivariate telemetry data to forecast tyre degradation, with methods such as XGBoost, LSTM, GRU, and Temporal Fusion Transformers. Although XGBoost achieves high accuracy and explainability, these models focus only on prediction and do not optimize pit stop decisions.

State-Space Modelling Approaches represent tyre degradation as a latent, time-varying process and model pit stops as state resets, effectively handling uncertainty and non-stationarity. However, they are limited to degradation modelling and lack real-time strategic optimization.

Strategy Optimization Techniques formulate pit stop planning as a search problem using algorithms like A*, enabling adaptive, real-time decision-making. Their limitation lies in relying on simplified or heuristic tyre degradation assumptions rather than learned predictions.

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Research Gaps

- Existing studies address **tyre degradation prediction**, **uncertainty-aware modelling**, and **pit stop strategy optimization** as separate problems rather than as a unified system.
- Predictive models achieve high accuracy but are **not integrated with real-time decision-making algorithms** for pit stop strategy.
- Strategy optimization approaches rely on **simplified or heuristic tyre degradation assumptions**, limiting their effectiveness under dynamic race conditions.
- Most solutions are **offline or research-focused**, with **no simple, interactive user interface** for simulation, testing, or real-time strategy evaluation.

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Research Objectives

To model and analyze **non-stationary tyre degradation behavior** in Formula 1 using advanced time series techniques.

To develop a **robust tyre performance forecasting model** capable of predicting future degradation trends under varying race conditions.

To identify **critical degradation phases (tyre cliff points)** that significantly influence pit stop timing and race strategy.

To design a **data-driven decision support framework** that assists in evaluating optimal pit stop windows during a race.

To integrate tyre degradation forecasts with **pit stop decision logic** for improved strategy planning.

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DATA FOUNDATION

Dataset Description

- **Sources:** FastF1 Python API datasets from the 2023-2025 seasons provide rich, granular data.
- **Collection Strategy:** Track-specific approach with 70% same-track history (3 years) + 30% similar tracks (2 years) matched by circuit type and temperature conditions
- **Key Features:** Lap time, tire life, stint progress, compound type, track conditions, degradation rate, plus 10 engineered features (moving averages, cumulative tire time, normalized metrics)
- **Scope:** 5 Drivers (VER, HAM, LEC, NOR, SAI) × 3 Compounds (SOFT, MEDIUM, HARD) × 5 Tracks (Bahrain, Monaco, Italy, Britain, Japan) = 75 models planned

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Proposed Work

1. **Input:** Driver code + Track name + Tire compound → **Output:** Recommended pit window (lap range)
2. **Baseline Model (ARIMA):** Univariate time series model using historical lap times to establish performance benchmark and validate methodology
3. **Enhanced Model (ARIMAX):** Incorporate exogenous variables (lap number for fuel load, stint progress) to separate tire degradation from fuel burn effects
4. **Advanced Model (XGBoost):** Non-linear machine learning approach using all 28 engineered features to capture complex degradation patterns
5. **Forecasting Strategy:** Generate 30-lap ahead predictions with 95% confidence intervals, identify critical degradation threshold (+2s from baseline), recommend pit window 2-3 laps before threshold
6. **Model Comparison:** Evaluate performance using AIC/BIC (information criteria), RMSE/MAE (forecast accuracy), and residual diagnostics (white noise validation)

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FINDINGS

Key Results and Discussion of Progress So Far

Dataset After Preprocessing for One Driver - Verstappen

When tested for a driver in the F1 race from historical data - Verstappen on Hard Tyres

• Raw data: 168 laps → Cleaned: 153 laps (8.9% removed)
• Coverage: 3 tracks (Bahrain, Abu Dhabi, Qatar), 3 years
• Outliers removed: 3 missing values, 3 slow laps, 5 out laps, 4 non-green flags
• Tire life range: 1-35 laps

Feature Engineering

• Original features: 18 columns → Final: 28 columns
• Engineered features: Lap time delta, degradation rate, cumulative tire time, normalized tire life, stint progress (%), moving averages (MA3, MA5), rolling std deviation

Key Finding

• Lap time trend: 96.89s (fresh tire, laps 1-5) → 88.57s (worn tire, laps 30+)
• Net change: -8.3s improvement (fuel effect dominates tire degradation)
• Implication: Univariate ARIMA captures combined effect; ARIMAX needed to isolate tire wear and gives us clarity based on other imporant variables

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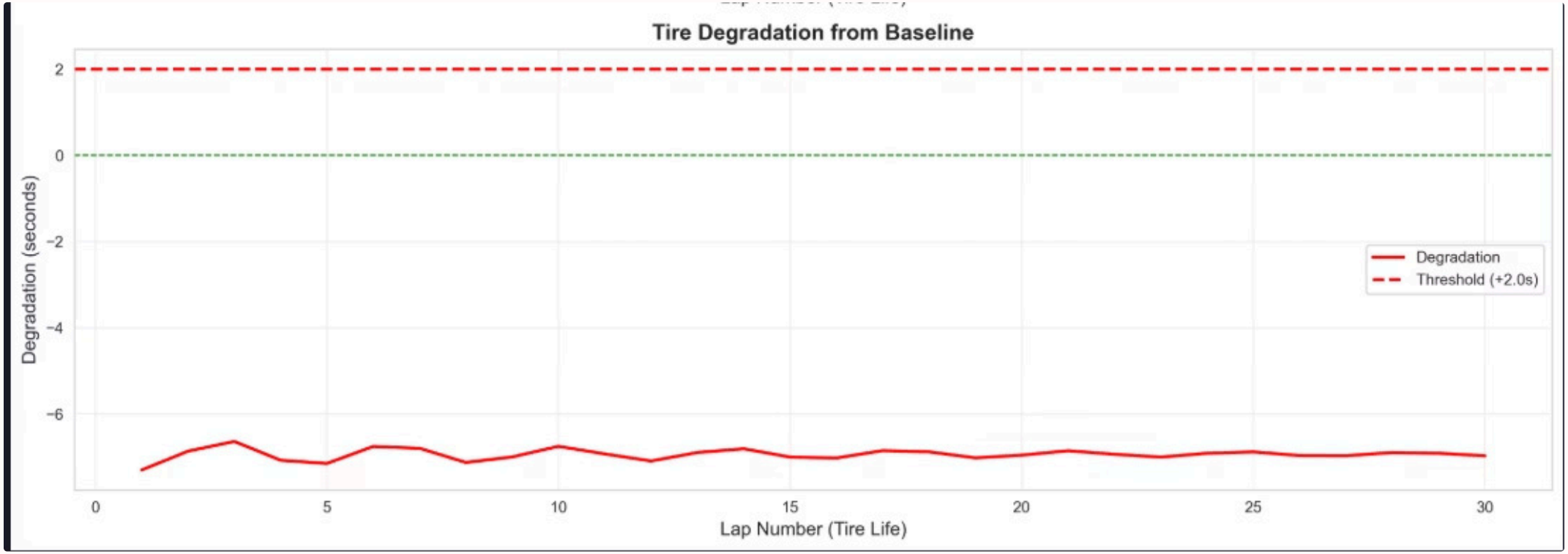

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Conclusion

• A track-specific data collection strategy was successfully implemented using FastF1 API, combining 70% same-track and 30% similar-track data matched by circuit type and temperature conditions.

• Comprehensive preprocessing pipeline reduced 168 raw laps to 153 clean samples with 10 engineered features including degradation rate, moving averages, and stint progress metrics.

• ARIMA(2,1,3) baseline model was developed following rigorous time series methodology.

• The model reveals fuel load effect dominates predictions (-7.3s improvement over 30 laps), confirming the need for ARIMAX with exogenous variables to isolate tire degradation.

• Future work includes ARIMAX implementation to control fuel effects, XGBoost for non-linear patterns, and scaling to 75 driver-compound-track combinations for comprehensive pit strategy prediction.

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Novelty of the Research

- Proposes a **data-driven tyre degradation forecasting approach** instead of relying on fixed or heuristic wear assumptions.
- Integrates **time series prediction** with **pit stop decision support**, bridging the gap between modelling and strategy.
- Focuses on identifying **tyre cliff points** to enable proactive and informed pit stop planning.
- Introduces a **simplified, user-accessible framework** for testing and evaluating pit stop strategies.

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Scope

- Models **Formula 1 tyre degradation** using time series data.
- Forecasts tyre performance to support **pit stop window analysis**.
- Identifies **tyre cliff points** affecting race strategy.
- Provides a **data-driven decision support framework** for strategy evaluation.

Limitations

- Dependent on **historical data quality and availability**.
- Dynamic race events (weather, safety cars) are **simplified**.
- Strategy support is **advisory**, not fully real-time or automated.

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